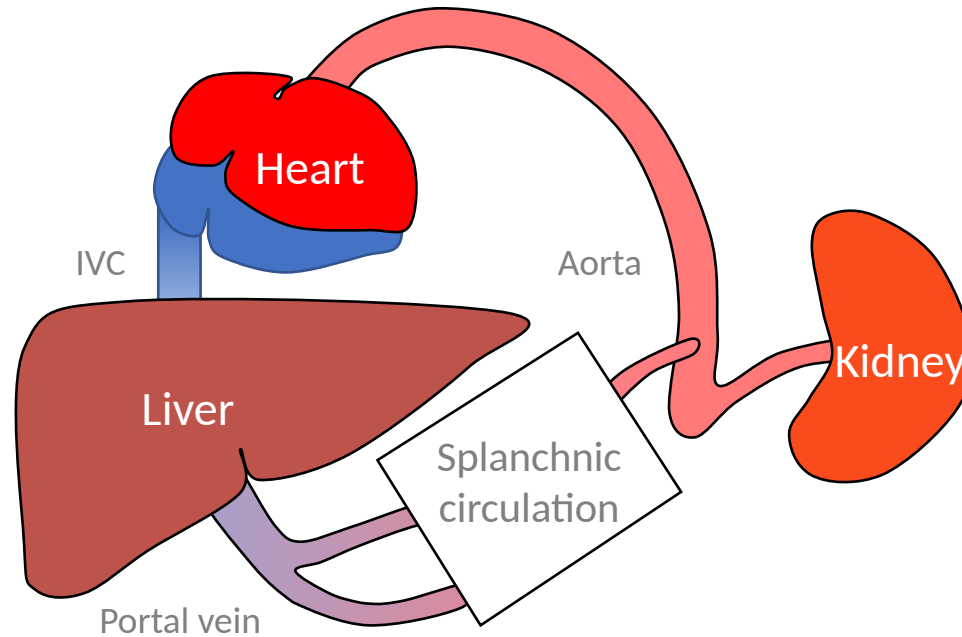


Differential Diagnosis of AKI in ESLD

Differential

Pathophys

Dx/Tx



AKI ↑ risk of death by 2-6X in patients with cirrhosis. Pre-renal causes account for > 60% of all AKI in patients with cirrhosis.

60-70%	PRERENAL	↓ volume (GI bleed, overdiuresis, GI losses) Sepsis (SBP, pneumonia, UTI) Medications HRS	Volume responsive Not volume responsive
30-40%	INTRARENAL	ATN AIN Glomerulonephritis	
<1%	POSTRENAL	Obstruction (clot, calculi, cancer, BPH, extrinsic compression) Neurogenic bladder	

Work-up to differentiate between causes should include:

- UA, urine electrolytes
- R/o bacterial infection (diagnostic paracentesis, blood and urine cultures)
- Assess liver function (e.g. RUQ + duplex)
- R/o GI bleed
- Consider renal US

Pathophysiology of Hepatorenal Syndrome

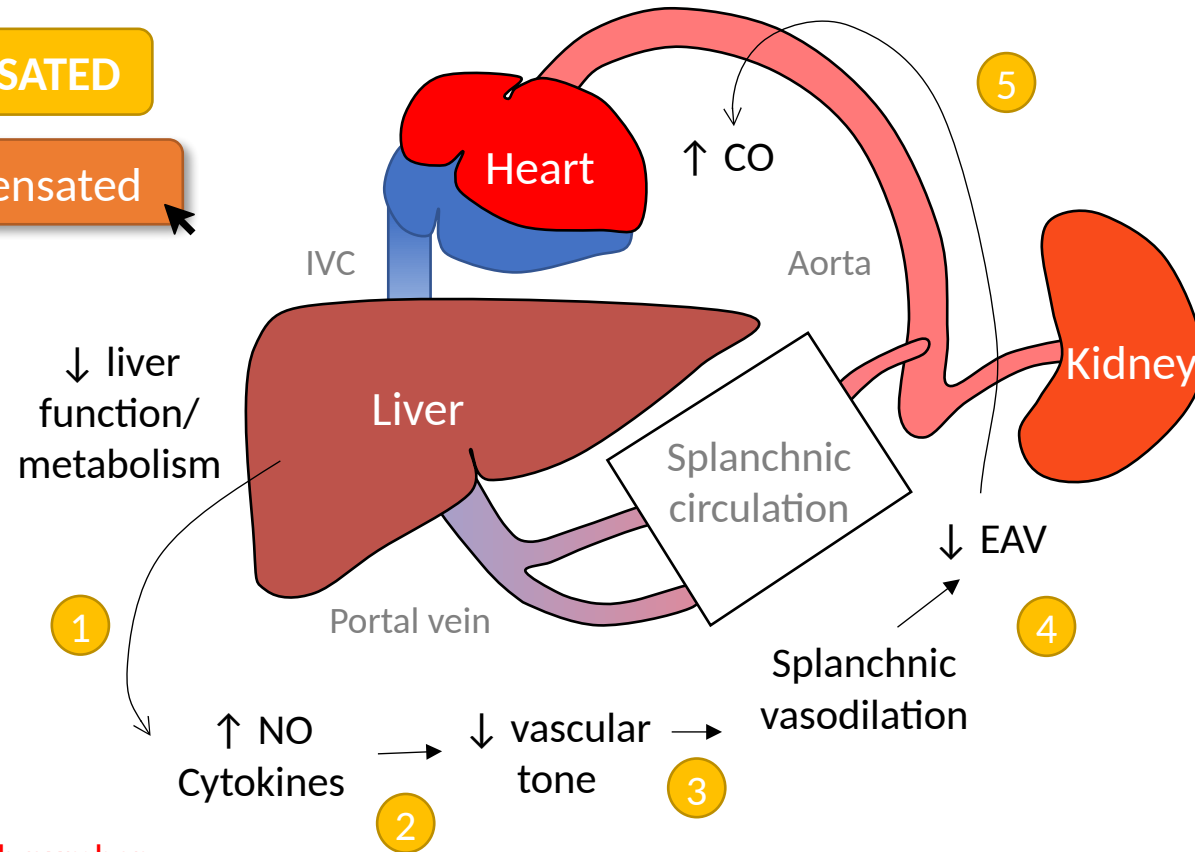
Differential

Pathophys

Dx/Tx

COMPENSATED

Decompensated



Click each number

In compensated cirrhosis, $\downarrow$ liver function leads to:

- 1 $\downarrow$ metabolism of vasodilatory substances which results in
- 2 $\downarrow$ vascular tone throughout the body including the splanchnic circulation (which supplies the gut)
- 3 Splanchnic vasodilation results in pooling of blood in the splanchnic circulation
- 4 This $\downarrow$ effective arterial volume (EAV)
- 5 The heart increases cardiac output in response to $\downarrow$ EAV which preserves blood pressure and organ perfusion

Pathophysiology of Hepatorenal Syndrome

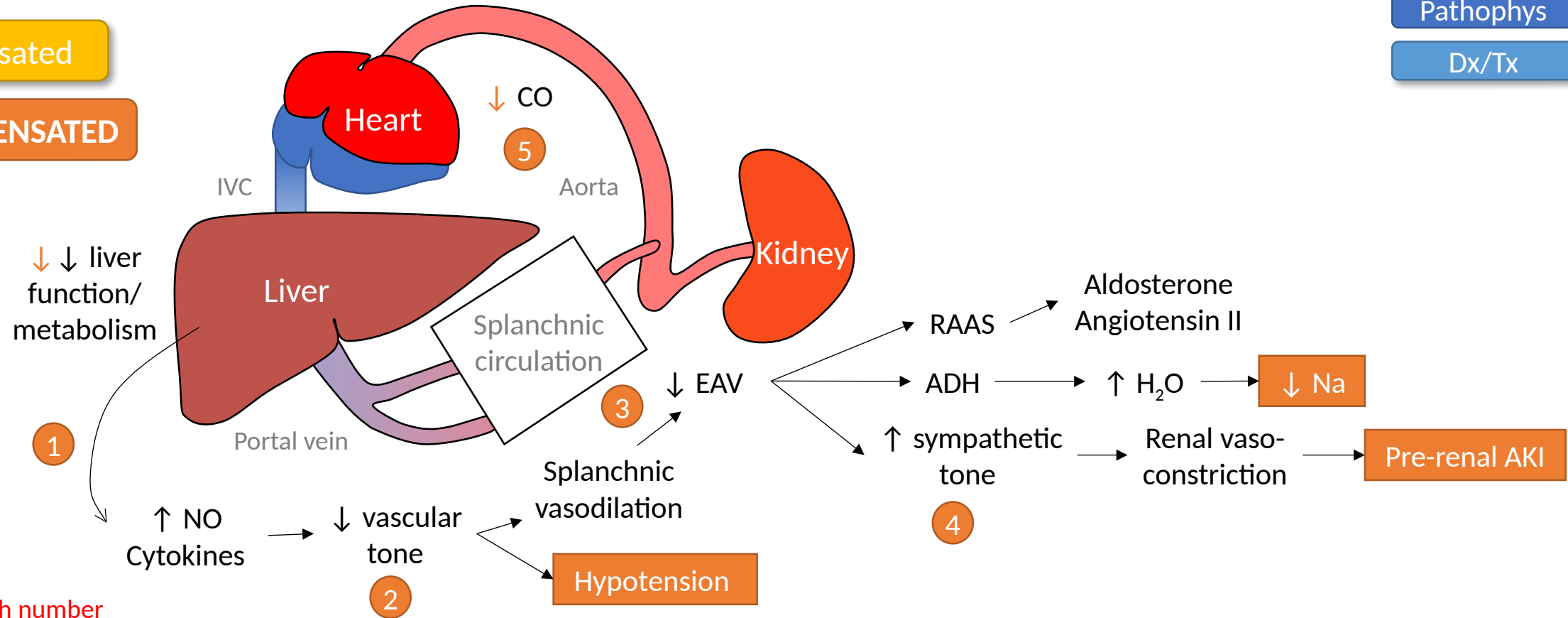
Differential

Pathophys

Dx/Tx

Compensated

DECOMPENSATED



Click each number

In DEcompensated cirrhosis, ↓ liver function leads to:

- 1** ↓ metabolism of vasodilatory substances which results in
- 2** ↓ vascular tone throughout the body including the splanchnic circulation
- 3** Splanchnic vasodilation results in pooling of blood in the splanchnic circulation, resulting in ↓ EAV
- 4** This ↓ EAV activates RAAS, ADH, and ↑ sympathetic tone leading to **hyponatremia** and **AKI**
- 5** Cardiac output fails to compensate leading to a ↓ in BP leading to pre-renal AKI and **hypotension**

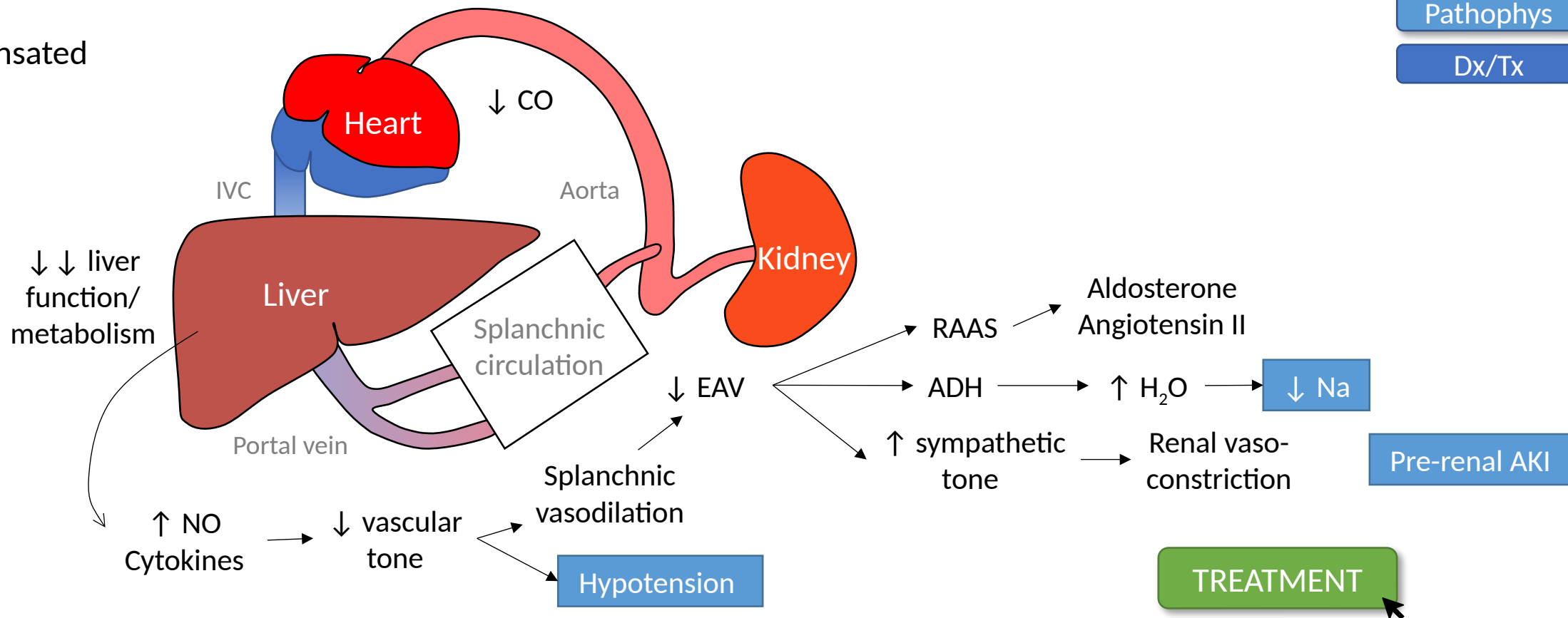
Diagnosis of Hepatorenal Syndrome

Differential

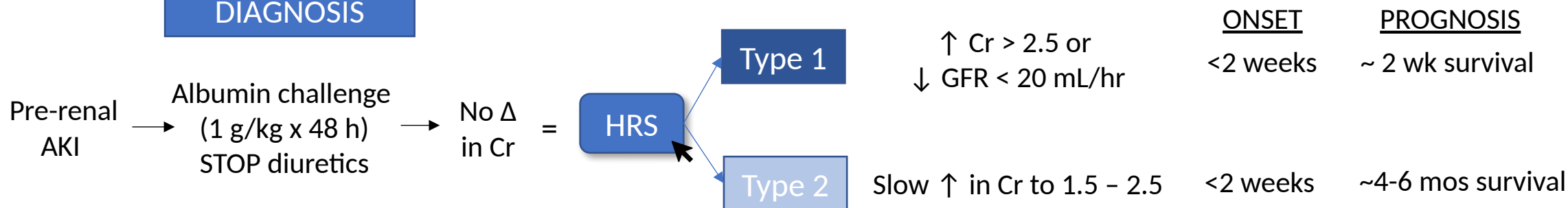
Pathophys

Dx/Tx

DECompensated
cirrhosis



DIAGNOSIS



Treatment of Hepatorenal Syndrome

Differential

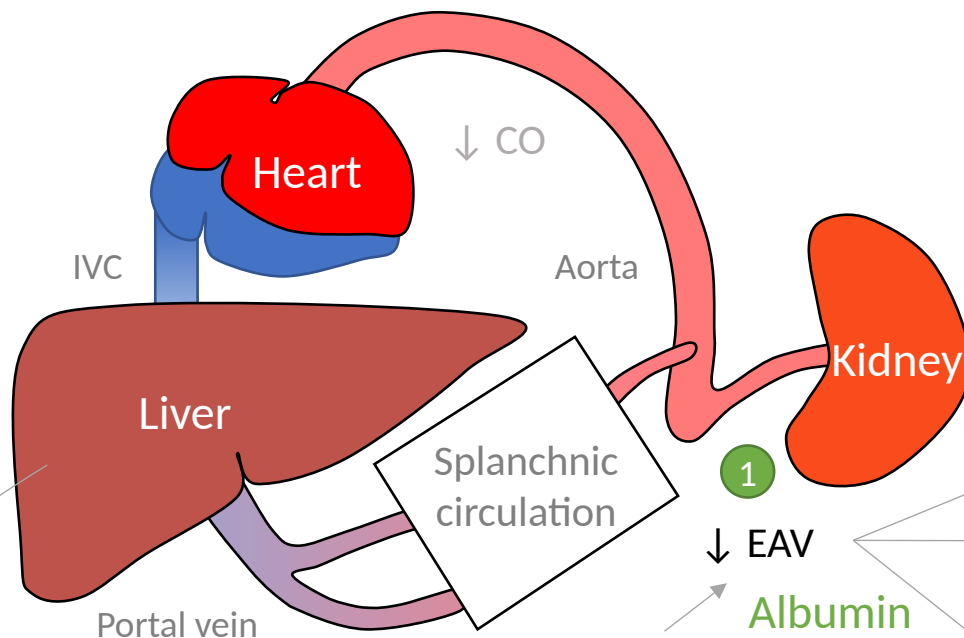
Pathophys

Dx/Tx

DECompensated
cirrhosis

TIPS
Transplant

4 ↓ ↓ liver
function/
metabolism



5 Dialysis

1

↓ EAV
Albumin

3 Splanchnic
vasodilation
Octreotide

RAAS

ADH

↑ sympathetic
tone

Aldosterone
Angiotensin II

↑ H₂O

Renal vaso-
constriction

↓ Na

Pre-renal AKI

↑ NO
Cytokines

2 ↓ vascular
tone
Vasopressors

Hypotension

DIAGNOSIS

Pre-renal
AKI

Albumin challenge
(1 g/kg x 48 h)
STOP diuretics

No Δ
in Cr

HRS

Type 1

Type 2

↑ Cr > 2.5 or
↓ GFR < 20 mL/hr

Slow ↑ in Cr to 1.5 - 2.5

Click each
number

TREATMENT

- 1 Albumin
- 2 Vasopressors (norepinephrine, midodrine) to ↑ MAP > 15
- 3 Octreotide - ↓ splanchnic dilation
- 4 TIPS
- 5 HD as bridge to liver transplant, which is ultimate treatment